

Untested Items

Area	What is not fully tested	Current evidence	Suggested verification	Priority
Workspace build	Full workspace build is only marked as partially implemented.	docs/test_plan.md lists setup as partially implemented.	Run <code>colcon build --packages-select my_tb3_world thermocator --cmake-args -DCMAKE_EXPORT_COMPILE_COMMANDS=ON</code> , then source <code>install/setup.bash</code> .	High
Gazebo simulation	Custom world launch, robot spawn, and /clock bridge still need live confirmation.	Gazebo launch is documented, but marked implemented and untested.	Run <code>ros2 launch my_tb3_world sim_with_bridge.launch.py use_sim_time:=true</code> and confirm Gazebo opens, robot spawns, and /clock publishes.	High
Simulated sensors	/scan, /odom, and /tf output from the simulated robot are not verified.	Test plan marks simulated movement and sensors as partially implemented and untested.	Echo /scan and /odom once while Gazebo is running; confirm TF tree is available.	High
DT mirror topics	/dt/* streams from dt_mediator are implemented but not runtime-tested.	dt_mediator exists and dt_integration.launch.py now starts the full DT pipeline.	Launch DT integration and check /dt/odom, /dt/scan, /dt/map, /dt/thermal_map, /dt/thermal_reading, /dt/robot_status, and /dt/environment_event.	High
Bidirectional command flow	/dt/cmd_vel -> /cmd_vel forwarding needs simulation verification.	Code forwards Twist to TwistStamped, but docs say motion must be verified.	Publish a test command on /dt/cmd_vel, echo /cmd_vel, and confirm the robot moves in Gazebo.	High
DT safety feedback	DT-side safety behavior must be verified with a mirrored obstacle event.	dt_safety_controller subscribes to /dt/environment_event and publishes a zero command to /dt/cmd_vel when an obstacle is nearby.	Put or simulate an obstacle near the robot, confirm /dt/environment_event reports OBSTACLE_NEARBY, /dt/control_status reports SAFETY_STOP, and /cmd_vel receives a zero command.	High
Internal state synchronization	New robot status mirroring must be verified in a live run.	status_monitor publishes /robot/status, and dt_mediator mirrors it to /dt/robot_status.	Echo /robot/status and /dt/robot_status, then confirm mode, sensor health, obstacle state, speed, and thermal health mirror correctly.	High
Synchronization tolerance	Sync delay has not been measured in a live run.	sync_monitor is implemented for odom, scan, thermal map, robot status, and environment event streams.	Run <code>dt_integration.launch.py sync_tolerance_seconds:=0.5</code> and confirm Sync ok or document warnings.	Medium
Decision node goal behavior	decision_node sending Nav2 goals is not scenario-tested with Nav2 active.	Technical status says goal behavior still needs scenario testing.	Start Gazebo, Nav2, thermal mapping, and decision node; verify goals are selected and accepted by Nav2.	High
Obstacle avoidance and propagation	Nav2 obstacle avoidance using /scan is partially implemented and obstacle events still need live mirroring verification.	status_monitor publishes /robot/environment_event, dt_mediator mirrors it, and dt_safety_controller reacts on the DT side.	Launch Nav2 with <code>nav2_thermal_params.yaml</code> , verify the robot plans around obstacles, and confirm obstacle state changes appear on /dt/environment_event and trigger /dt/control_status.	High
Object interaction simplification	Object interaction is documented as a simplified equivalent rather than a fully tested physical interaction.	Test plan marks it as a limitation.	Demonstrate hotspot detection plus simulated deployment event at a selected waypoint.	Medium
End-to-end scenario	The full demo path is documented but not proven repeatable from a clean start.	Demo checklist exists, but scenario testing is untested.	Follow <code>docs/demo_checklist.md</code> from clean terminals and record pass/fail notes.	High
RViz thermal display plugin	Custom thermocator/ThermalDisplay plugin is a known crash risk.	docs/demo_checklist.md recommends avoiding it in the safe demo path.	Either debug the plugin separately or keep it out of the final demo.	Medium
Thermal grid math/indexing	Previous thermal grid math/indexing issues are documented as known risks.	docs/technical_status.md says these were not fixed.	Validate thermal map coordinates against known robot positions and hotspot locations.	Medium
Nav2 thermal costmap plugin	Registered/configured plugin still needs runtime verification.	Technical status says it is untested and may need lifecycle or parameter adjustment.	Launch Nav2 with the configured plugin and inspect costmaps/planning behavior.	Medium

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Gazebo bridge topic names	Topic names may differ if the active TurtleBot3 Gazebo model uses namespaced topics.	Technical status flags possible topic-name adjustment.	Run ros2 topic list and compare with src/my_tb3_world/config/ros_gz_bridge.yaml.	Medium